



BUYER'S GUIDE

40 Graphics Cards



Motherboards

A facilitator, the motherboard is one hot category this year... New chipsets, features, and better value than ever before!

A motherboard is one component which you'd not like to change for your processor's tenure. Its important to think of a motherboard as more than a hunk of PCB that connects your CPU, GPU, and memory. Your choice of motherboard dictates upgradeability, stability, extent of overclocking, and, to a lesser degree, performance.

A motherboard is never the first preference while shopping though. All of us prefer to CPU-shop first and then decide on a board. Be that as it may, don't neglect selection of this component and let your vendor play God with your cash.

Today's motherboards are every bit as personal as a cell phone, and vendors have spent a lot of money designing boards that actually have a number of feature add-ons. Imagine having a motherboard with a remote control that lets you control all the important functions—like shutting down, hibernating, playing and controlling multimedia content—movies and music.



What You Should Be Looking At

Storage: Besides compatibility for at least a future iteration of processors, motherboards will always have a certain amount of expansion, internally and externally. Internally, look



Agent Tips

Don't allocate a budget for a motherboard after splurging on a CPU and RAM. Instead decide on a budget for the board at the beginning... and stick to it!

for the number of SATA ports and also the number of PATA ports. Four of the former should suffice, though if you already have three or four HDDs and plan on a SATA DVD burner, then look for a board that has six ports—the latter in case you've got a couple of working PATA drives with you.

Memory: Look for four memory banks. Two reasons for this. One—memory is dirt cheap and 4 GB is a possibility now, and will be a necessity later. Two—single 2 GB DIMMs are very costly.

Some vendors will try to sway you saying "This board supports 8 GB or 16 GB". While their claims will be true you need to see how much memory you are actually going to install which will be based on what you're using the board for. The support for extra memory will be there on 99 percent of all motherboards—you needn't worry about it.

Chipset: A motherboard is never more than the sum of its Northbridge and Southbridge. Take note in particular of the Northbridge (as the Southbridge is usually paired with it). You should know the basic differences between common chipsets and more importantly the features and the age of the chipset. This ensures future compatibility. This is where consumer education becomes important. Remember an informed decision is one you will hardly ever regret.

Connectivity options: Look for the number of USB ports on the rear (four is good; some newer boards have six, which is better). This is especially useful since most devices like thumb drives, printers, memory card readers,

game controllers all use USB connects. We recommend at least four USB ports, six would be great, and if your motherboard supports extra ports via add-in cards—all the better. If you are looking at external drives and such, look for a FireWire port and E-SATA ports (which most high-end boards come with). A few boards may also come with Wireless access points inbuilt. If you have a notebook or Wi-Fi-enabled PDA, this is a definite plus as you'll be able to surf the Internet or even transfer data to and fro without



having to worry about investing in a wireless access point or router—this is a definite plus.

Enthusiast Features

Look for Reset, Power and CMOS clear buttons on the board—a real boon when dealing with freeze-ups due to overclocking or general instability. Also look for POST debuggers—very useful for (duh!) debugging during boot-up problems. Overclockers will also want to look at BIOS options, which determines the extent of overclocking shenanigans possible. Layout of motherboards is also important for overclockers as you will probably look at better cooling solutions such as aftermarket CPU coolers, or even water blocks and liquid cooling solutions. ■